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Submitted
by

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Assignment

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Submitted
To

Mr - Iftikhar
Ali

Q1 [CLO-1]

A Spooling

Spooling refers to the process of managing input/output (I/O) operations by temporarily storing data on a disk or in memory before it is sent to target device (like a printer).

Difference from buffering

Spooling involve storing data on a storage (like a disk) while buffering uses a temporary memory area (buffer) to hold data during its transfer between devices.

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B Role of Spooler in managing Devices like printer.

=> Queuing Print Jobs
it store and organizes Print jobs in a queue... So that each Job is sent to the printer one at a time in the correct order

=> Error handling
if printer become unavailable, the Spooler holds Jobs until the issue is resolved.

=> efficient Resource utilization
it allows the CPU and application to continue working without waiting for the printer to finish each Job.

C Advantage

⇒ Improved efficiency
The CPU and I/O devices can work simultaneously, leading to better utilization of system resource

⇒ Job management
It allows multiple jobs to be queued and processed in an orderly manner without manual intervention

Disadvantage

⇒ Storage Requirement
Spooling requires additional storage spaces to hold the queued data, which may not always be available

⇒ Complexity
Implementing and managing
a Spooling system can
add Complexity to the
operating system.

**A Total time taken to
complete all 3 Jobs.**

The printer handles
one room at a
Job

time, so the Jobs
are completed sequentially
in the order they are
submitted (J1 → J2 → J3)
lets calculate time for
each job

J1: 20 pages at 1 page
per second → 20 seconds

J2: 10 pages at 2 pages
per second → $10 \text{ pages} \div 2 \text{ pages/sec}$
= 5 second

J3: 15 pages at 1.5 pages
per second → $15 \text{ pages} \div 1.5 \text{ pages/sec}$
= 10 Sec

Total time = 35

b Timeline of Job Start and Finish Times

- $\Rightarrow$ 1: Starts at 0 Seconds and Finishes at 20 sec
- $\Rightarrow$ 2: Starts at 20 Seconds and Finishes at $20+5=25$ sec
- $\Rightarrow$ 3: Starts at 25 sec and Finishes at $25+10=35$ sec

Time line

0 - 20 Seconds: J1

20 - 25 Seconds: J2

25 - 35 Seconds: J3

c How Spooling Improves Efficiency

Spooling significantly enhance efficiency in the printing process by allowing jobs to be submitted and queued without waiting for the printer to become available. Here's how

=> ~~Parallel processing~~ ~~multiple printer~~
Process one Job, the
Spooler allows the
System to accept new
Jobs and Queue them,
enabling users or applications
to continue working without
delay.

=> orderly execution
Spooling ensures that
Jobs are executed in
the correct order without
Conflicts or interruption.

=> ~~2010: good~~ ~~Printer~~ ~~with~~
User Convenience ~~2010: 1979~~
Users can Submit
their jobs and move on
to other tasks instead
of waiting for each Job
to complete.